

# goxpyriment — Getting Started

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## Getting Started with goxpyriment

Welcome! If you are a psychologist or neuroscientist used to building experiments in Python (with **Expyriment** or **PsychoPy**), you are in the right place.

goxpyriment is a alternative framework to build experiments using the **Go** programming language, offering significant advantages in timing precision and ease of sharing your work.

This document provides a few examples for you to get acquainted with the framework (if you want to run the examples, you will need goxpyriment installed and a working Go toolchain. See [Installation](#) for instructions).

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### Tutorial 1: Your First Trial

A classic sequence: **Fixation Cross** (500 ms) → **Target Circle** → **Wait for Spacebar**.

```
package main
```

```
import (  
    "log"  
  
    . "github.com/chrplr/goxpyriment/control"  
    "github.com/chrplr/goxpyriment/stimuli"  
)
```

```

func main() {
    exp := NewExperimentFromFlags("SimpleTrial", Black, White, 32)
    defer exp.End()

    fixation := stimuli.NewFixCross(20, 3, control.White)
    target    := stimuli.NewCircle(50, control.White)

    exp.ShowInstructions("Press SPACE when you see the circle.")

    exp.ShowTimed(fixation, 500) // hold fixation for 500 ms

    exp.Show(target)
    exp.Keyboard.WaitKey(K_SPACE)
}

```

The following example goes a bit further: it runs a loop where a white square and a short tone are presented every second.

```

package main

import (
    . "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := NewExperimentFromFlags("Test", Black, White, 32)
    defer exp.End()

    w, h, _ := exp.Screen.Size()
    rect := stimuli.NewRectangle(0, 0, float32(w)/2.0, float32(h)/2.0, White)

    sound := stimuli.NewTone(440, 120, 0.5)
    sound.PreloadDevice(exp.AudioDevice) // generates the sound

    instr := stimuli.NewTextBox(
        "A white rectangle will be displayed periodically\n\nPress any key to start\n\n",
        600, FPoint{}, White)

    exp.AddDataVariableNames([]string{"onset", "duration"})

    exp.Run(func() error {
        exp.Show(instr)
        exp.Keyboard.Wait()

        for {
            onsetNS, _ := exp.ShowTS(rect)
            sound.Play()

```

```

        exp.Wait(200)
        exp.Screen.Clear()
        offsetNS, _ := exp.Screen.FlipTS()
        duration_ms := (offsetNS - onsetNS) / 1_000_000
        exp.Wait(800)

        exp.Data.Add(onsetNS/1_000_000, duration_ms)
    }
})
}

```

---

## Tutorial 2: Blocks, Trials, and Data Logging

For real research you need multiple trials and a CSV result file.

```

package main

import (
    "fmt"
    "log"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/design"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("ParityTask", control.Black, control.White, 3)
    defer exp.End()

    exp.AddDataVariableNames([]string{"number", "is_even", "rt_ms", "correct"})

    err := exp.Run(func() error {
        exp.ShowInstructions(
            "Is the number EVEN or ODD?\n\nPress F for Even, J for Odd.",
        )

        numbers := []int{1, 2, 3, 4, 5, 6, 7, 8}
        trials := design.MakeMultipliedShuffledList(numbers, 4) // 32 trials, randomized

        for _, n := range trials {
            exp.Blank(1000)

            stim := stimuli.NewTextLine(fmt.Sprintf("%d", n), 0, 0, control.White)
            exp.Show(stim)
        }
    })
    if err != nil {
        log.Fatal(err)
    }
}

```

```

        // WaitKeysRT returns the key pressed, the reaction time in ms, and any error
        key, rt, _ := exp.Keyboard.WaitKeysRT(
            []control.Keycode{control.K_F, control.K_J}, -1,
        )

        isEven := (n % 2 == 0)
        correct := (isEven && key == control.K_F) || (!isEven && key == control.K_J)

        // Subject ID and timestamp are prepended automatically.
        exp.Data.Add(n, isEven, rt, correct)
    }
    return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
}

```

`exp.Run()` handles the detection of the ESC keypress to interrupt the experiment. Two files are written to `~/goxpy_data/` automatically when the experiment ends:

- `<expName>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>.csv` — pure CSV data rows, directly importable by Excel, R, or pandas.
- `<expName>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>-info.txt` — session metadata (start/end time, hostname, OS, framework version, display and audio configuration, participant info).

---

## Tutorial 3: Rapid Serial Visual Presentation (RSVP Streams)

Many paradigms — RSVP, Attentional Blink, priming — need stimuli flashed at high speed with precise timing. `goxpyriment` has dedicated stream functions that:

- Disable GC during the stream (no garbage-collection pauses).
- Lock every onset and offset to a VSYNC boundary.
- Record a `TimingLog` (predicted vs. actual onset) for every stimulus.
- Capture any key or mouse event that occurs during the stream.

### Text stream

The simplest entry point is `stimuli.PresentStreamOfText`:

```

package main

import (
    "fmt"
    "log"
    "time"

```

```

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("RSVP Demo", control.Black, control.White, 360)
    defer exp.End()

    exp.AddDataVariableNames([]string{"target_pos", "detected", "rt_ms"})

    // A stream of words; "TIGER" is the target at position 3 (0-indexed).
    words      := []string{"CHAIR", "RIVER", "LAMP", "TIGER", "CLOCK", "STONE", "BREAD", "TIGER"}
    targetPos  := 3

    err := exp.Run(func() error {
        exp.ShowInstructions(
            "Words will flash rapidly on screen.\n\n" +
            "Press SPACE as quickly as possible when you see TIGER.\n\n" +
            "Press any key to start.",
        )

        // 200 ms on, 50 ms off per word → ~4 words per second
        on  := 200 * time.Millisecond
        off := 50 * time.Millisecond

        events, logs, err := stimuli.PresentStreamOfText(
            exp.Screen, words, on, off,
            0, 0, // centred on screen
            control.White,
        )
        if err != nil {
            return err
        }

        // Did the participant press SPACE during the stream?
        ev, detected := stimuli.FirstKeyPress(events, control.K_SPACE)
        rtMs := ev.Timestamp.Milliseconds()

        // Optional: inspect timing quality. Read the authoritative SDL-clock
        // fields (OnsetNS/OffsetNS), not the Go-clock ActualOnset/ActualOffset
        // diagnostics. Achieved on-screen duration vs the target is the jitter.
        for i, l := range logs {
            shownMs := int64(l.OffsetNS - l.OnsetNS) / 1_000_000
            jitterMs := shownMs - l.TargetOn.Milliseconds()
            fmt.Printf("word %d: target %d ms shown %d ms jitter %d ms\n",
                i, l.TargetOn.Milliseconds(), shownMs, jitterMs)
        }
    })
}

```

```

    exp.Data.Add(targetPos, detected, rtMs)
    return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
}

```

Similar functions exist to present streams of images or sounds:

- Images:

```

stims := []stimuli.VisualStimulus{pic1, pic2, fixation, pic3}
on    := 100 * time.Millisecond
off   := 0 * time.Millisecond

elements := stimuli.MakeRegularVisualStream(stims, on, off)
events, logs, err := stimuli.PresentStreamOfImages(exp.Screen, elements, 0, 0)

```

- Sounds

```

tones      := []stimuli.AudioPlayable{tone440, tone880, tone440}
onsets     := []int{0, 500, 1000} // ms from stream start
durations  := []int{200, 200, 200}

elements, _ := stimuli.MakeSoundStream(tones, onsets, durations)
events, logs, err := stimuli.PlayStreamOfSounds(elements)

```

All stream functions return (`[]UserEvent`, `[]TimingLog`, `error`) to analyse timing quality and log participant responses.

---

## Tutorial 4: Hardware-Precision RT with Event Timestamps

`WaitKeysRT` measures reaction time from the moment the function is called. That works well for a single-stimulus trial, but breaks down when several stimuli appear in sequence and you need RT from a specific onset.

Consider a **masked priming** paradigm: a prime word appears briefly, then a target appears 500 ms later, and you want RT measured from the prime onset — not from when `GetKeyEventTS` was called. With the standard approach you would need to record a timestamp before the prime and do arithmetic afterward. The event-timestamp API handles this directly.

SDL3 stamps every keyboard event with a hardware-interrupt time (`KeyboardEvent.Timestamp`, nanoseconds). `exp.ShowTS(stim)` returns the SDL nanosecond time captured immediately after the VSYNC flip. Because both values are on the same clock, their difference is hardware-precision RT — no arithmetic needed:

```

package main

import (
    "fmt"
    "log"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("PrimingRT", control.Black, control.White, 3600)
    defer exp.End()

    exp.AddDataVariableNames([]string{"prime", "target", "key", "rt_prime_ns", "rt_target_ns"})

    primes := []string{"DOCTOR", "NURSE", "BREAD", "TABLE"}
    targets := []string{"NURSE", "DOCTOR", "TABLE", "BREAD"} // related / unrelated pairs
    responseKeys := []control.Keycode{control.K_F, control.K_J}

    err := exp.Run(func() error {
        exp.ShowInstructions(
            "A word will flash, then a second word will appear.\n\n" +
            "F = Living thing    J = Non-living thing\n\n" +
            "Respond to the SECOND word as quickly as possible.\n\n" +
            "Press SPACE to start.",
        )

        for i := range primes {
            exp.Blank(500) // inter-trial interval

            // 1. Show prime – record its VSYNC flip timestamp
            prime := stimuli.NewTextLine(primes[i], 0, 0, control.Gray)
            primeOnset, _ := exp.ShowTS(prime)
            prime.Unload()

            exp.Wait(500) // prime–target SOA

            // 2. Show target – record its VSYNC flip timestamp
            target := stimuli.NewTextLine(targets[i], 0, 0, control.White)
            targetOnset, _ := exp.ShowTS(target)
            target.Unload()

            // 3. Wait for response – get hardware event timestamp
            key, eventTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, 3000)

            // 4. Compute RTs from each stimulus onset

```

```

    rtFromPrime := int64(eventTS - primeOnset) // nanoseconds
    rtFromTarget := int64(eventTS - targetOnset) // nanoseconds

    exp.Data.Add(primes[i], targets[i], key, rtFromPrime, rtFromTarget)
    fmt.Printf("prime RT: %.1f ms    target RT: %.1f ms\n",
        float64(rtFromPrime)/1e6, float64(rtFromTarget)/1e6)
}
return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
}

```

Key observations:

- `exp.ShowTS(stim)` does the same thing as `exp.Show(stim)` — clear → draw → flip — and additionally returns the nanosecond timestamp of the flip.
- `GetKeyEventTS` returns the SDL3 event timestamp (not a polling delta), so subtracting any previously recorded `ShowTS` onset gives a physically meaningful RT.
- Both timestamps are in SDL nanoseconds (divide by `1e6` for milliseconds). Storing raw nanoseconds in the data file and converting offline is the recommended practice.
- `GetPressEventTS` provides the same capability for mouse responses.

---

## Tutorial 5 — Test AI Coding

1. Take the PDF of a scientific paper about a psychology experiment and ask Gemini or Claude to read it and generate a detailed description (stimuli, design, timings) to give to an AI coder to replicate it. Save the answer in a `description.md` file.
  2. Open Gemini or Claude (command-line tool), and ask it to implement the experiment detailed in `description.md`, using the `goxpyriment` library (provide the path of your local clone of the repository) and taking inspiration from the examples provided in the `examples/` folder.
- 

## Next Steps

- Browse the `examples/` folder for complete, documented paradigms (Stroop, Simon, Posner, Sperling, QUEST threshold estimation, and more).
- Read the [User Manual](#) for a deeper explanation of every concept, or the [API Reference](#) for complete function signatures.
- Report bugs and suggestions at <https://github.com/chrplr/goxpyriment/issues>.
- Happy experimenting!